

MOLLER POLARIMETER TEST at 11 GeV

18-20 December, 2015

CONGRATULATIONS!

12 GeV in CEBAF

11 GeV in the Hall A

NEW POLARIZED ELECTRON TARGET



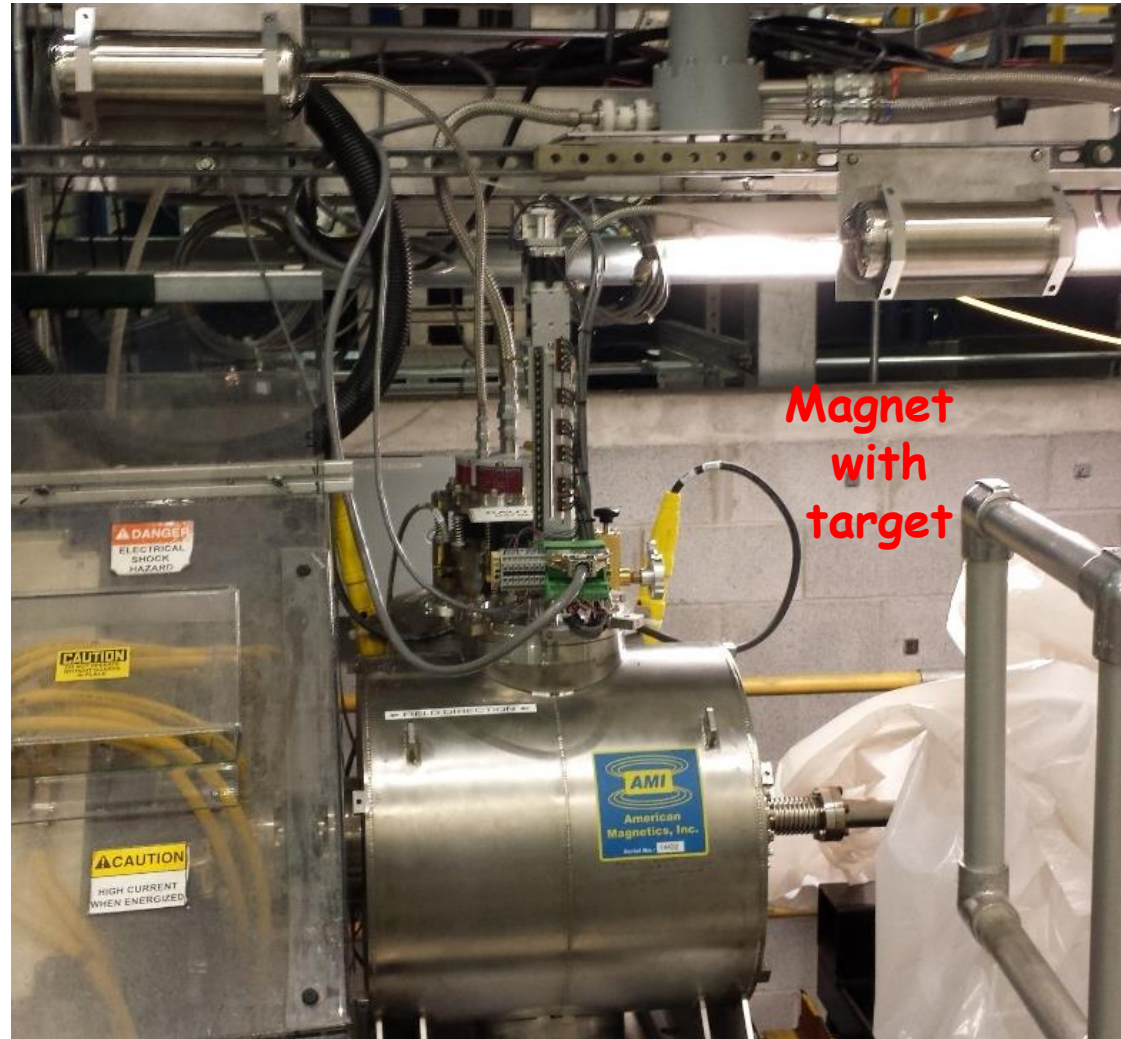
Power supply



Compressor



Water chiller



Magnet with target

Superconductive magnet:

Maximal Field $\pm 5T$

Very limited information

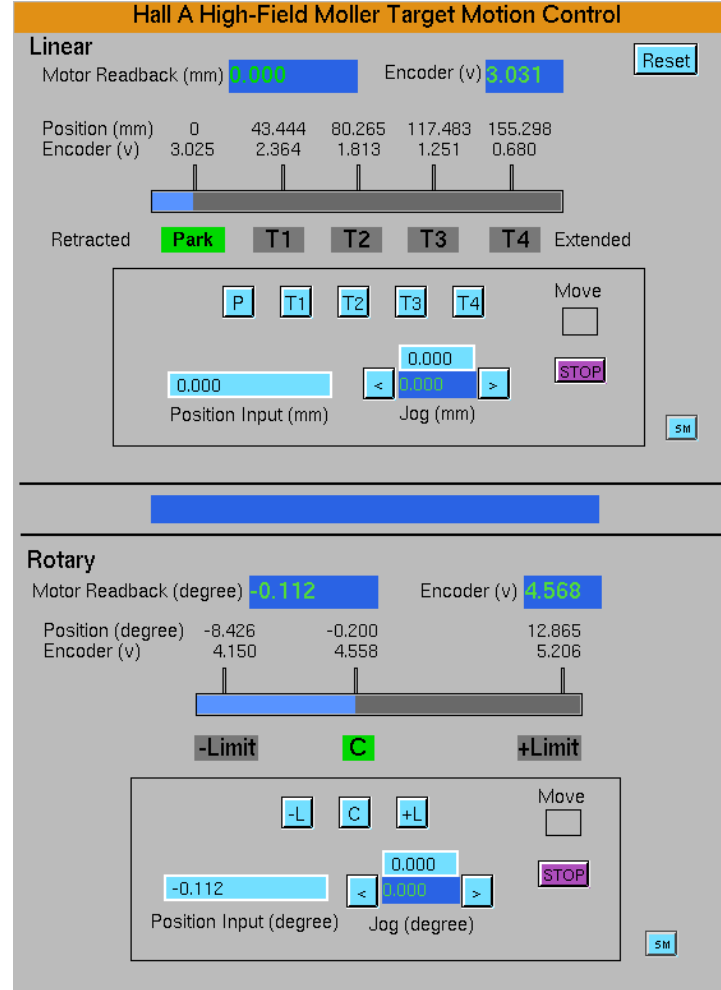
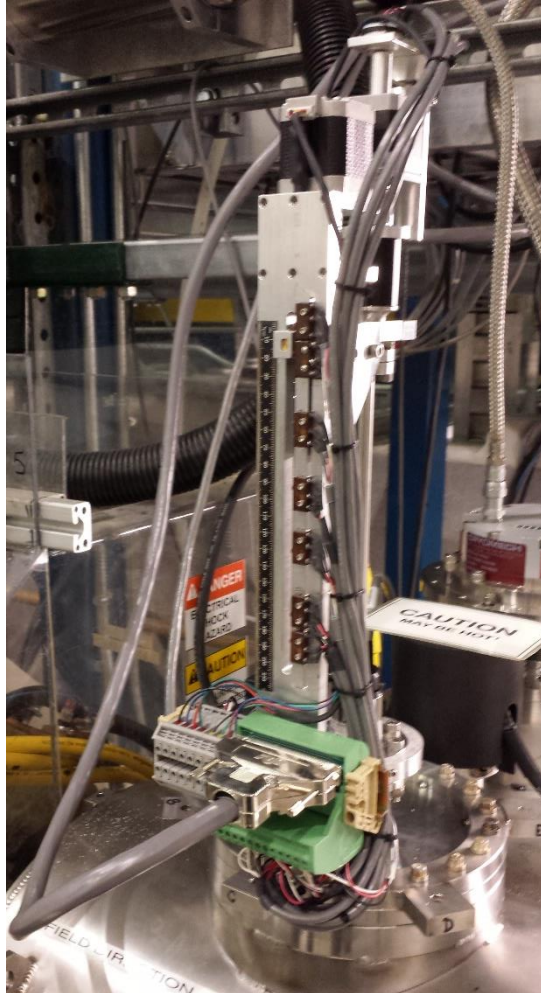
- portable*
- no LN required*
- no cryo lines required*

Small amount of He ->

low thermal capacity and "cold" production ->

low field ramping gradient ($\sim 35\text{min}$: $0 \rightarrow 3T$)

NEW POLARIZED ELECTRON TARGET



Made by Temple University
Rotation in horizontal plane $\pm 10^\circ$
Vertical translation

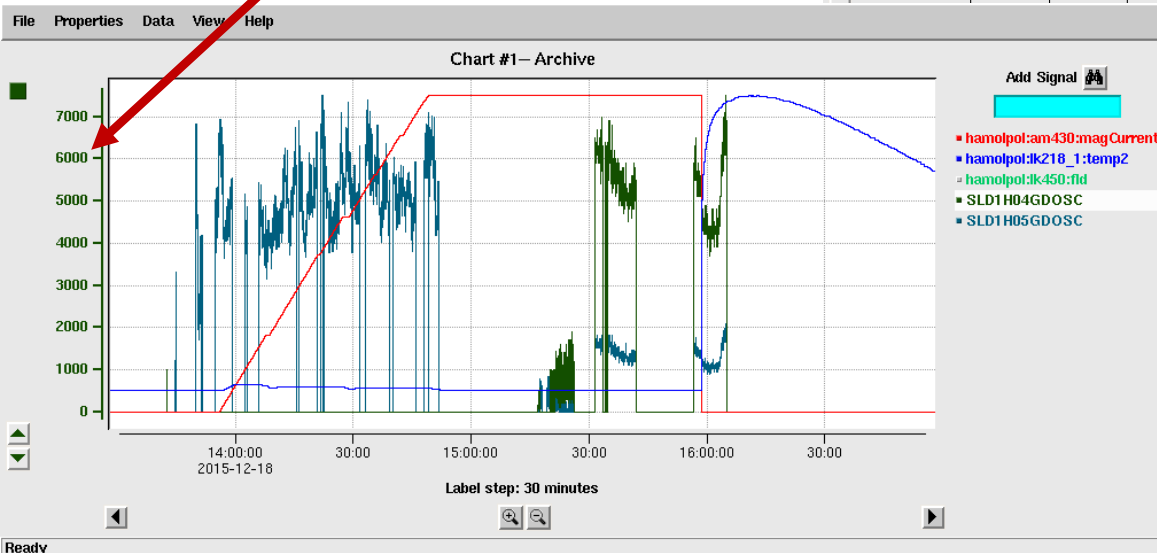
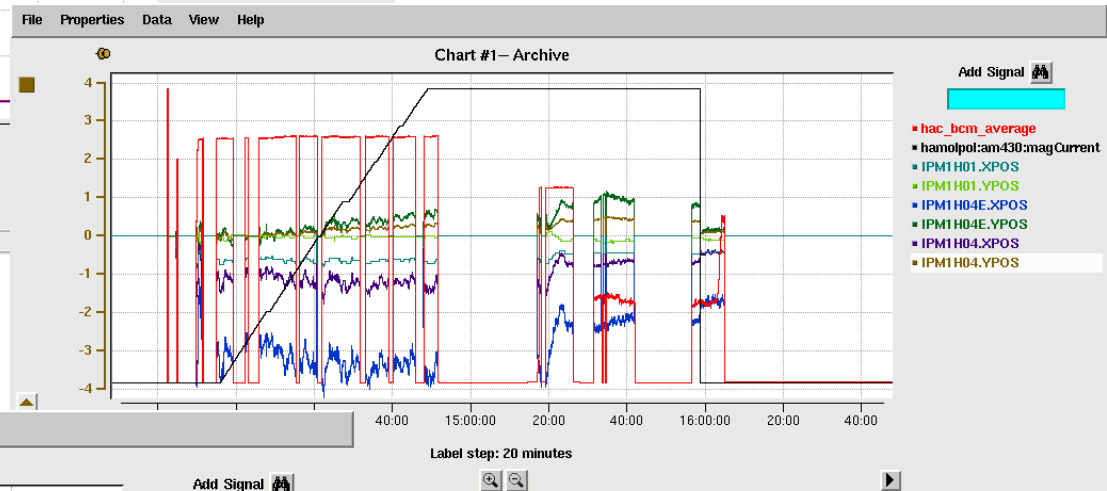
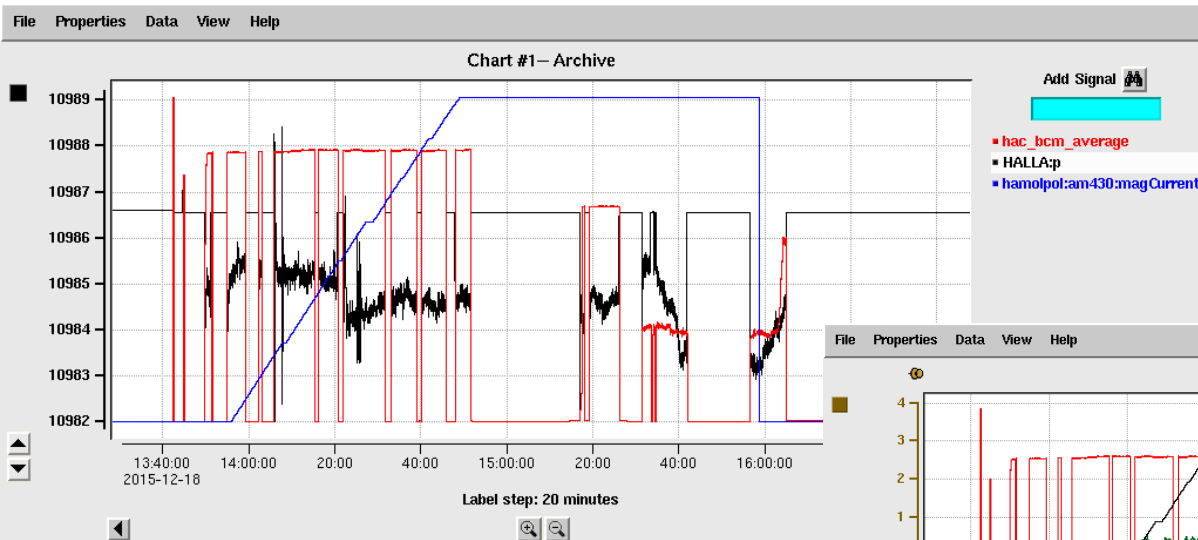
Targets in holder: pure iron $1\mu\text{m}$, $4\mu\text{m}$, $12\mu\text{m}$, $25\mu\text{m}$

PLAN FOR THE TEST

- *To test new SC magnet with the beam*
- *To test the target assembly with the beam*
- *To test the Moller polarimeter with 11GeV*

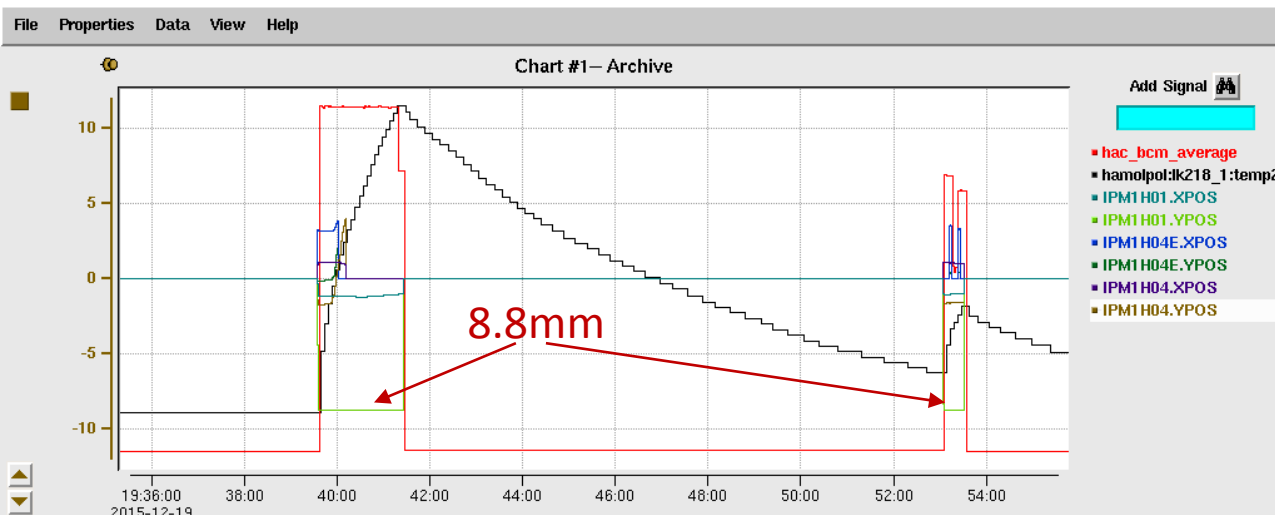
18 December, 2015

CW beam 1.5μA
Straight beam
4T field in magnet
Target #3 (12μm)
*Magnet **QUENCH***

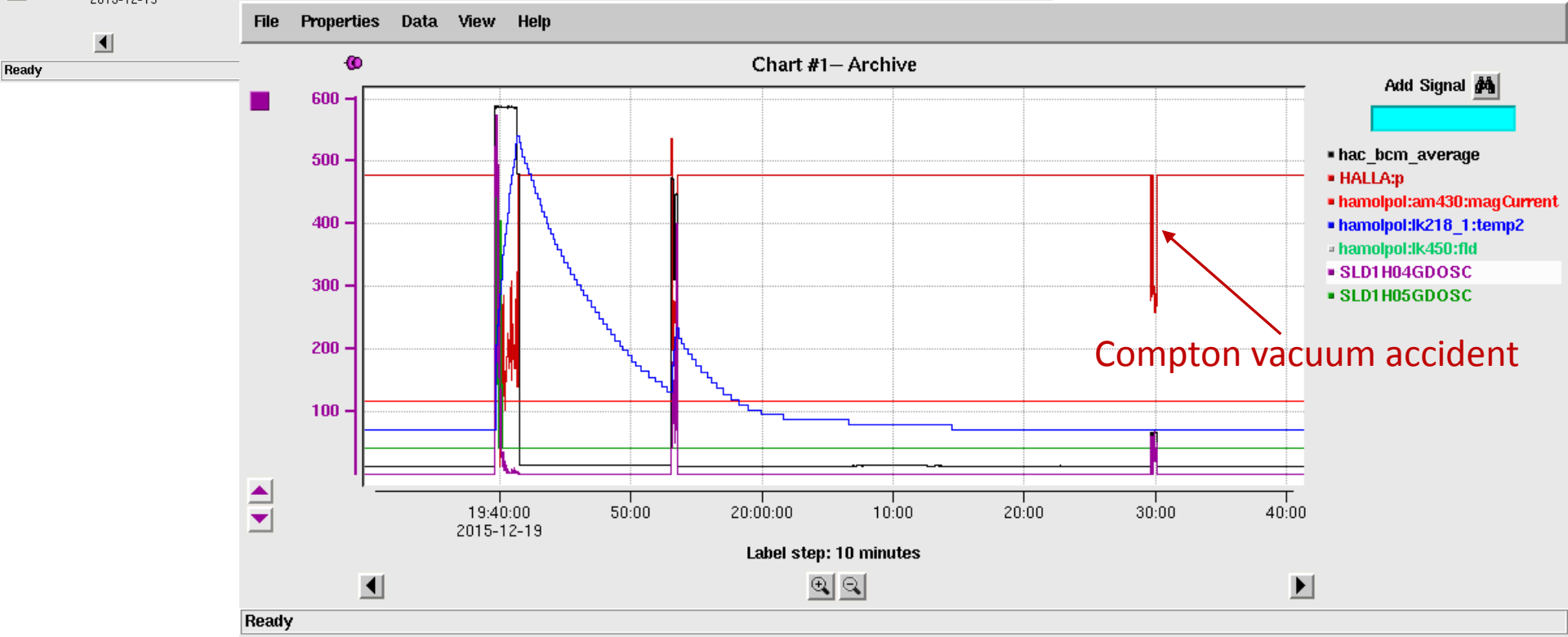


**5-6 hours for
the magnet recovery
after quench!**

19 December, 2015



*Beam through Compton
Pulse/Tune
Low radiation
No field in magnet
No Moller target*



20 December, 2015

2am: MCC set the Moller settings
ramp the Moller magnet up to
3T

Tune beam/no beam

6:30am - **CW!** Taking data

Target #2 (4um), 0.5uA

Target #3 (12um), 1-1.2uA

7:47am - ramp magnet up to **4T**

Target #3 (12um), 4T, 1.1uA

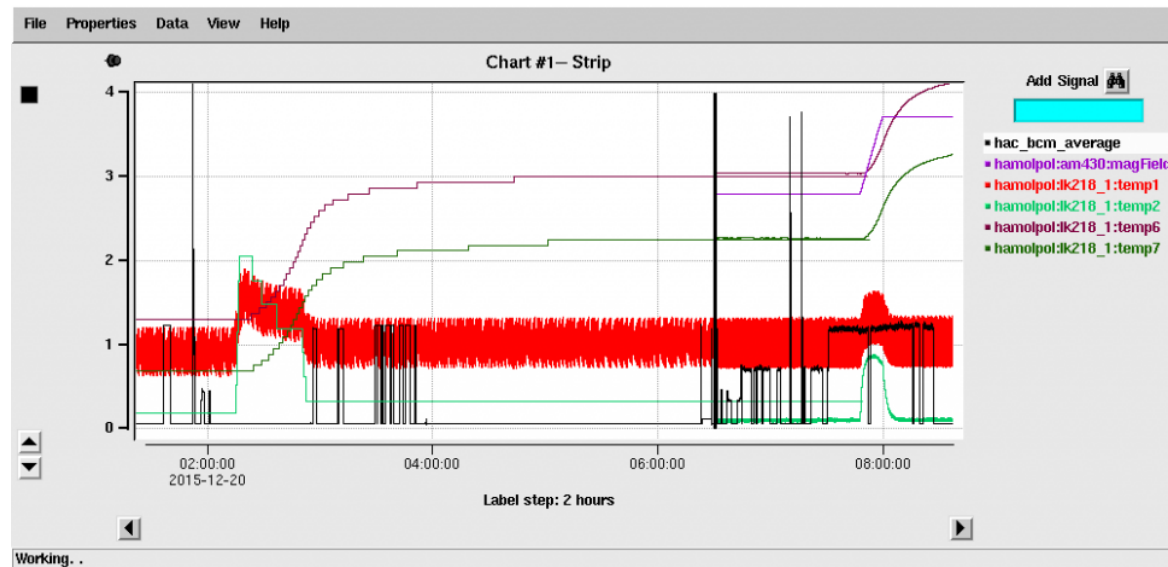
Very preliminary

Beam polarization Target #3:

at 3T: -56.91 ± 0.22

at 4T: -59.43 ± 0.22

Straight beam
Low radiation
Good beam tuning
~2 hours of CW

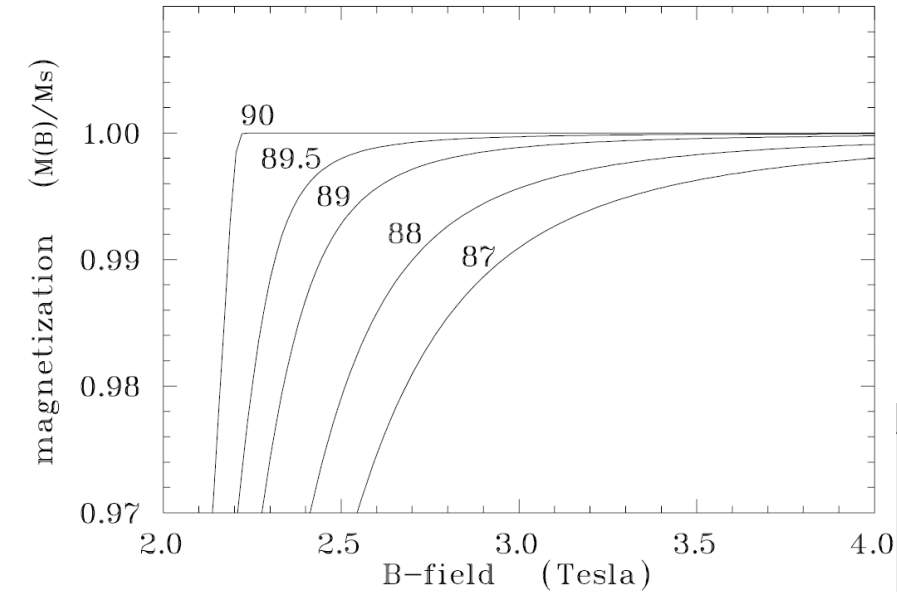


Low polarization:

- Did not tune detector (HV, thresholds)
- Unknown target angle -> target is not saturated -> low polarization
- **Unprecise energy** -> unknown precession angle
- Unprecise Wien filter

Need spin dance and target angle tuning

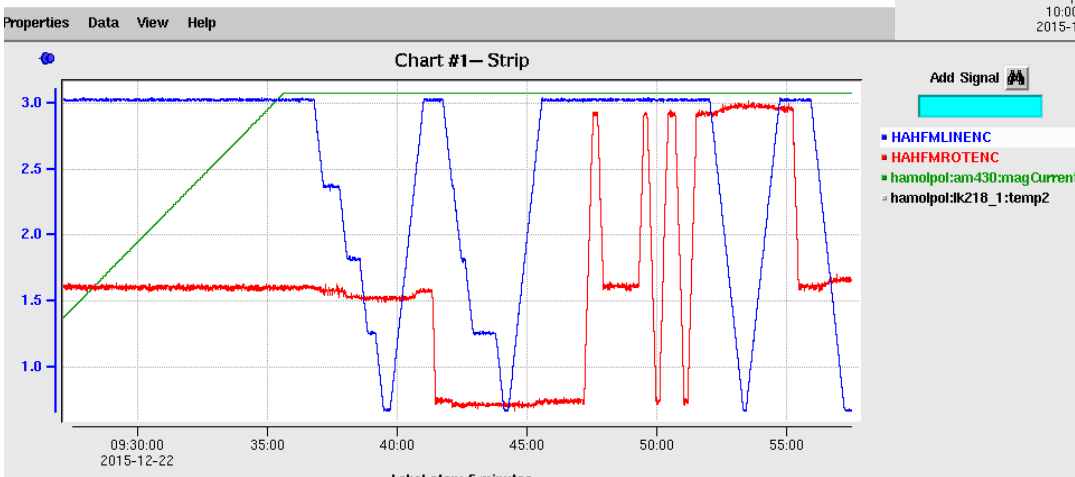
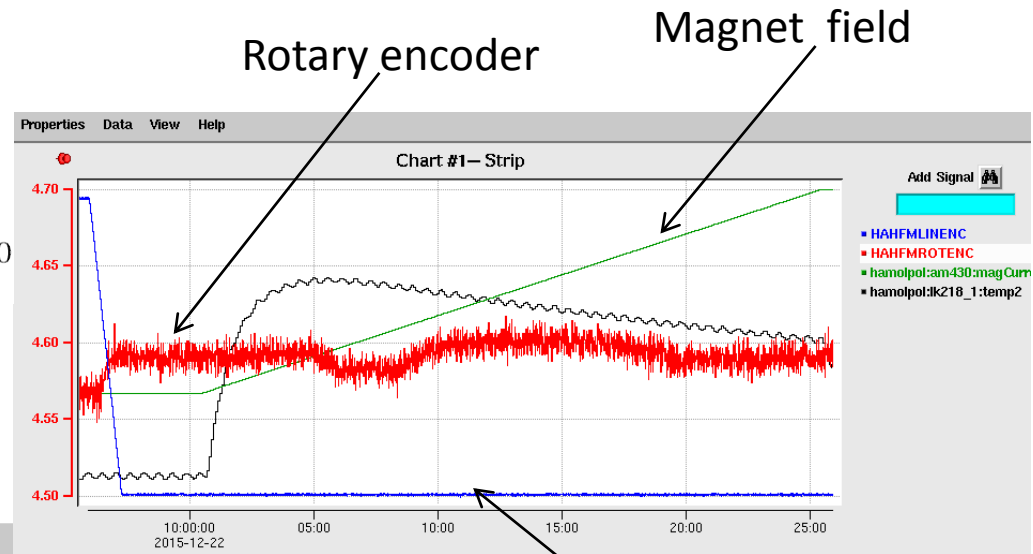
Target Rotation Test



*Magnetic coupling?
Motor in different
magnetic field?*

Goal: yaw + pitch $< 0.5^\circ \rightarrow 3T$

Target rotation angle dependence of
- magnet field
- target position



Linear encoder

*Target angle:
Accuracy?
Repeatability?*

SUMMARY

Moller polarimeter at 11GeV:

- *Optics (quads, dipole, power supplies) OK*
- *Detector OK*
- *Moller DAQs (old and FADC) OK*
- *New SC magnet:*
 - ~2hours of CW, quench, lost of superconductivity*
 - need more experience*
 - magnet alignment check at low beam energy*
 - Don't shoot the magnet!***
 - beam tuning requirements*
 - coexisting with Compton?*
- *New target*
 - have to discuss the data with Temple*
 - need more data (target orientation)*